

Transient gain with adaptation frozen vs active: what the first measurement says

2026-09-13, R5611351, Claude Code session 90c5825b-..., repo FractionalReservoir, branch main, HEAD e0810e4. Data: data/transient_gain_smoke (fast mode: $n = 500$, one seed, $T = 20$ s, six regular states per regime, horizon 1 s, 11 min). Figures: figs/transient_gain_smoke. Code: run_transient_gain, SRNNCellTypePairs.transient_gain, fig_transient_gain, fig_transient_gain_excursions. Background: Non_normal_amplification.md (the measure and why the dendritic block), Session_2026-09-12_..._handoff.md (the discussion that led here).

1. What was measured

At a sampled state S_0 of the noisy trajectory, propagate the n dendritic unit vectors through the tangent equation and read the x-in / x-out block of the propagator every 20 ms:

$$G(t) = \| P_x \Phi(t) P_x^T \|_2 \quad (\text{worst case over dendritic directions})$$

plus, from the same $n \times n$ block, the noise-average gain $\|\cdot\|_F/\sqrt{n}$, the gain along the E/I difference mode, the E/I sum mode and the leading Lyapunov direction at that state, and the top right singular vector at the peak (the optimal input pattern). Three propagators Φ per state:

name	Φ	what it is
J _{xx} frozen	$e^{\{J_{xx} t\}}$	the dendritic block fixed at S_0 : the conventional rate-network linearisation, adaptation and depression held constant
J frozen	$e^{\{J t\}}$	the full Jacobian fixed at S_0 : a, b, g respond linearly, the state does not move
active	Φ from $J(S(t))$ along the stored trajectory	what a tangent perturbation actually does

Adjacent pairs isolate one effect each: J_{xx} frozen $\rightarrow$ J frozen is adaptation's linear feedback at a fixed operating point; J frozen $\rightarrow$ active is nonstationarity (the state moves). $G(0) = 1$ by construction; the propagator is verified against expm to $2e-6$ on small networks.

2. Numbers (1-s horizon, medians over six states)

regime	λ_1	J _{xx} frozen	J frozen	active	$e^{\{\lambda_1 \cdot 1 \text{ s}\}}$
no adaptation	+4.0	47 000	47 000	1 200	55
single-timescale (sfa1_std1)	+0.44	11 000	143	30	1.6
multiple-timescale (sfa3_std2)	-0.11	39	6.7	8.6	0.9

The same numbers as ratios, which is how to read them:

regime	J _{xx} frozen / J frozen	J frozen / active	active / $e^{\{\lambda_1 t\}}$
no adaptation	1 (no a, b states: $N = n$)	39	≈ 22
single-timescale	78	4.8	≈ 20
multiple-timescale	5.8	0.78	≈ 10

Direction readings (active propagator, at the peak): the optimal input pattern has a participation ratio of ~ 60 of 500 neurons, E fraction ~ 0.5 , and $|\cos| \leq 0.15$ with the leading Lyapunov direction, in every regime. The E/I difference

mode is the best of the named directions in every regime; the E/I sum mode is the worst and falls below 1 once adaptation is on; the noise-average gain is ~ 1 in the two adapted regimes.

Peak times: the two frozen propagators and the active one in the two unstable regimes peak AT THE HORIZON ($t_{\text{peak}} = 1.0$ s), i.e. they are still growing. Only the multiple-timescale active curve has a genuine peak (0.18 s) followed by decay.

3. How to understand it

3.1 The frozen picture is a runaway, not a transient, wherever $\alpha(J_{xx}) > 0$

In the no- and single-timescale regimes the dendritic block at a typical state has spectral abscissa $\alpha \approx +7$ and $+6$ s^{-1} (fig_transient_amplification). Its propagator $e^{J_{xx} t}$ therefore grows like $e^{\alpha t}$ times a non-normal prefactor without bound; the “peak gain” is just the gain at the horizon. The $\omega - \alpha$ margin of the earlier figure is the initial slope of that curve. So the static rate-network view does not describe a transient at all in those regimes: it describes exponential divergence at a frozen operating point that the real system never sits at. Comparisons involving the frozen propagators must be made at a fixed t , not at “the peak”.

3.2 Adaptation’s linear feedback at the operating point is large

Holding the state fixed but letting a and b respond (J frozen) cuts the 1-s gain by $78\times$ in the single-timescale regime and $5.8\times$ in the multiple-timescale one. Mechanism: a dendritic perturbation δx raises δr , which raises δa on τ_a and depletes δb on $\tau_{\text{rec}}/\tau_{\text{rel}}$, and the effective input $x - c \cdot \Sigma a$ and the synaptic output $r \cdot \Pi b$ both drop. This is the effect the instantaneous ω cannot see (it is $t = 0^+$), and it is the cleanest statement that adaptation belongs in the conceptual model of the recurrent dynamics: same weights, same state, two orders of magnitude less amplification in a second.

Why the reduction is smaller in the multiple-timescale regime ($5.8\times$ vs $78\times$): its J_{xx} frozen gain is already small (39 vs 11 000) because the operating point differs – STD has depressed W_{eff} and SFA has moved neurons to a shallower part of ϕ , the $\omega = 46$ vs 74 story – so there is less left for the feedback to remove within 1 s. The 10-s SFA rung has not acted within this horizon at all.

3.3 Nonstationarity is a second, separate reduction, present even without adaptation

Without adaptation the two frozen propagators are identical (a sanity check of the code: $N = n$), yet the active gain is $39\times$ smaller than the frozen one. The linearisation at any single point of a chaotic trajectory has $\alpha \approx +7$, but the trajectory diverges only at $\lambda_1 = +4$: the unstable directions rotate as the state moves, and no direction stays aligned with the local worst case for long. Freezing the Jacobian overstates amplification in every regime, adapted or not. In the single-timescale regime the J -frozen curve’s late log-slope is about $+3$ s^{-1} against $\lambda_1 = +0.44$: the trajectory is more stable than any of the states along it, which is what “stabilisation through motion” looks like when adaptation keeps moving the operating point.

3.4 What actually amplifies: 10-20 $\times$ along a specific pattern

The honest quantity is the active gain divided by $e^{\lambda_1 t}$: the transient amplification over and above asymptotic divergence or decay. It is 22, 20 and 10 in the three regimes. That is the Murphy-Miller / Hennequin balanced-amplification quantity for this network, and adaptation reduces it only by about $2\times$; the four-orders-of-magnitude spread in the frozen column is mostly the change in λ_1 ($+4 \rightarrow -0.1$) and in the operating point, not a change in non-normality. In the stable regime the whole $8.6\times$ is a transient: peak at 0.18 s, then decay, the textbook shape. In the other two regimes the transient rides on exponential divergence and never peaks within a second.

3.5 The amplifying pattern is not the chaotic direction

The optimal input pattern is a ~ 60 -neuron E/I-imbalanced pattern with $|\cos| \leq 0.15$ to the leading Lyapunov direction (whose x part we compared; the full leading vector lives $\sim 80\%$ in the SFA states in the adapted regimes). Two different mechanisms, then: chaos expands along a slow, adaptation-dominated direction; transient amplification is a fast dendritic E/I pattern. The E/I difference mode being the best named direction everywhere confirms the balanced-amplification prediction; the E/I sum mode being damped (< 1) in the adapted regimes says a uniform

push engages every neuron's adaptation and is cancelled – relevant to the stimulation idea (Next 3), where the prior was that the sum mode is the direction that engages adaptation without a large transient.

The noise-average gain ~ 1 in the adapted regimes means the network's own additive noise is not amplified on average; only structured perturbations are, 10-30 $\times$ along the optimal pattern. Adaptation makes the network selectively sensitive rather than globally quiet.

3.6 Active can exceed J frozen (the stable regime: 8.6 vs 6.7)

The plan expected the frozen gains to bound the active one; they do not. The local Lyapunov rate fluctuates around λ_1 (positive about a third of the time in this regime, in excursions of a few hundred ms), and a moving state passes through stretches with more expansion than the sampled point had; time-varying linear systems can amplify more than any of their frozen coefficients allow. The same ordering appeared on the 40- and 60-neuron test networks. With six states from one seed the 8.6 vs 6.7 difference is inside the state-to-state spread, so it is a caution ("do not call the frozen gains bounds; report all three"), not yet a result.

4. What this means for the paper

- The conventional rate-network linearisation (J_{xx}) overstates amplification by orders of magnitude in the adapted regimes, and by 40 $\times$ even without adaptation. Say so, with the three-curve figure.
- Adaptation's contribution at the operating point is the J_{xx} -frozen vs J-frozen ratio (78 $\times$ and 5.8 $\times$ at 1 s). Its contribution to the asymptotic rate is λ_1 (+4 $\rightarrow$ +0.44 $\rightarrow$ -0.11). Its contribution to the non-normal prefactor is modest (22 $\rightarrow$ 20 $\rightarrow$ 10). Keep these three apart; the frozen picture conflates them.
- The number to quote for "how much does a dendritic perturbation grow before adaptation reverses it" is the active peak in the stable regime: 8.6 $\times$ at 0.18 s (one seed; get the medium-mode value). In the unstable regimes quote the gain at a fixed t together with $e^{\lambda_1 t}$.
- The amplifying pattern is an E/I difference pattern over ~ 60 neurons, distinct from the Lyapunov direction; the E/I sum pattern is damped.
- Keep the $\omega - \alpha$ figure as the conventional view, captioned as the initial slope of the frozen curve.

5. Caveats

One seed, six states, 1-s horizon. Tangent-space linearisation along a noisy trajectory: G is an upper envelope of real amplification (saturation ignored). The horizon is too short for the 10-s SFA rung. The excursion sampling (Next 2) found only 0-2 onset / quiet states per regime in 20 s; the onset-vs-quiet contrast is untested.

6. What to do next

1. Run the stage at medium ($T = 40$ s, 2 seeds, 12 regular + up to 8 onset
 - 8 quiet, horizon 2 s, ~ 1 h): `run_transient_gain('run_mode','medium','out_dir', full-file('data','topk_med','transient_gain'))`, then the two figures with `run_dir = data/topk_med`. This gives the resolved peak in the stable regime, the seed spread, and the first real onset-vs-quiet contrast.
2. Record the spectral abscissa of the FULL J at every sampled state (a few eigs per state, or the late log-slope of the J-frozen curve) next to λ_1 : "the trajectory is more stable than any of its states" is a claim worth a number.
3. A 3-s horizon on fewer states, so the 10-s rung begins to act and the single-timescale curve can be seen to peak or not.
4. Next 3 (stimulation directions): the E/I sum mode's damping supports the prior; the intermittent-but-stable networks are listed by `find_intermittent_stable` (multiple-timescale regime at $\mu_{IE_relative}$ 2.06 or $\mu_{EE_relative}$ 13.4-15.7; single-timescale at $n = 190$).